AI 500 - Practice Problem Set 5

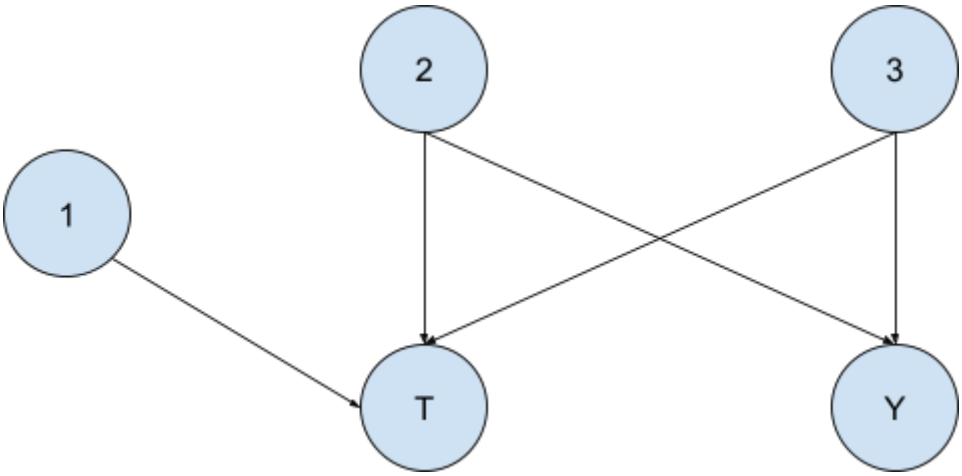
Q1.

A single comparison of outcomes of the same group before and after administering the treatment is known as:

A. Randomized Control Trial
B. Cross sectional comparison
C. Difference-in-differences comparison
D. None of the above

Q2.

What is (are) the confounding variable(s) in the following DAG?



A. 1
B. 2
C. 3
D. None of the above

Q3.

Which of the following is **NOT** a quasi-experimental method?

A. Randomized Controlled Trial (RCT)
B. Difference-in-Differences
C. Regression Discontinuity Design
D. Instrumental Variables

Q4.

The “Parallel Trends” assumption is crucial for which method?

A. Pre-post
B. Simple difference
C. Difference-in-Differences
D. Instrumental Variables

Q5.

Suppose the model is $Y = 3X + 5T + \epsilon$. What is the Average Treatment Effect (ATE)?

Q6.

Suppose you have the following data. Compute DID.

Group	Before	After
Treatment	40	65
Control	50	70

Q7.

A researcher studies the relationship between diabetes and gallbladder disease among hospital patients. In the general population, these two diseases occur independently. However, both

increase the chance of hospitalization

Disease Status	Probability in General Population	Probability of Hospitalization (Given Condition)
Diabetes only	5%	30%
Gallbladder disease only	5%	30%
Both diseases	1%	60%
Neither disease	89%	5%

(a) Are diabetes and gallbladder disease independent in the population?

(b) Explain what happens to their relationship when we only look at hospitalized patients.

(c) Why does this happen? Illustrate using causal graph reasoning (collider logic).

Q8.

A study evaluates whether a new drug (T=1) improves recovery (Y=1). Data are recorded by gender and treatment:

Treatment: Drug (T=1)

Gender	Recovered	Not Recovered	Total	Recovery rate
Men	81	6	87	?
Women	192	71	263	?
All	**273**	**77**	**350**	?

Control: No Drug (T=0)

Gender	Recovered	Not Recovered	Total	Recovery rate
Men	234	36	270	?
Women	55	25	80	?
All	**289**	**61**	**350**	?

(a) Compute ATE
(b) Compute CATE by gender
(c) Compute a weighted adjusted ATE by giving each gender equal weight
(d) What do you observe?

Q9: You are given the following causal diagram:

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Z → T → Y

Z → Y
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(a) Is the causal effect of T on Y identifiable from observed data?
(b) What variable(s) should be controlled for to identify the causal effect?
(c) Write the expression for the identified causal effect using the adjustment formula.

Q10. A study compares the recovery rate between two hospitals (A and B):

Hospital	Treated (%)	Recovered (%)
A	80	70
B	20	90

However, when data are pooled across both hospitals, the overall recovery rate is *higher* for Hospital A.

(a) Explain how this reversal can occur.
(b) Which variable acts as a confounder here?
(c) How would you represent this situation using a causal DAG?
(d) What does this paradox teach us about conditioning and aggregation in causal inference?

Q11. A hospital records data on two conditions among admitted patients: asthma and pneumonia.
In the general population, having asthma does not affect the chance of having pneumonia.

However, because the hospital tends to admit patients who have either asthma or pneumonia (or both), the data among *admitted* patients looks like this:

Asthma	Pneumonia	Number of Admitted Patients
Yes	Yes	10
Yes	No	40
No	Yes	40
No	No	10

(a) In the admitted sample, do asthma and pneumonia appear to be positively or negatively related?
(b) Are they actually related in the general population?
(c) Explain why this apparent negative relationship arises when we only look at admitted patients.